Utku Demirtaş

Software Engineer

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SUMMARY

Backend-focused software engineer building real-time engines, event-driven services and RAG pipelines, plus the CI/CD and containers that ship them. Computer engineering graduate working across Node/TypeScript, Spring Boot/Java and Python.

EXPERIENCE

Founder & Lead Developer *Okut Gitsin — okutgitsin.com (launching soon) | Nov 2025 – Present*

- Designed and built a live-stream vehicle auction platform end to end, from data model to deployment
- Engineered an atomic bid engine on Redis and Lua for millisecond-level, race-free bidding
- Built real-time bidding over WebSocket and live video auctions with LiveKit
- Added an escrow payment flow (held, refunded, forfeited) with IBAN verification, Helmet CSP/HSTS, rate limiting and GitHub Actions CI/CD

AI Trainer (Contract) *Mercor | Aug 2026 – Nov 2026*

- Scored, compared and rewrote frontier-LLM responses for text-optimization tasks against rubrics I authored
- Wrote prompts designed to surface failures in reasoning, accuracy and instruction-following

DevOps & Backend Developer *Extramus | Jul 2025 – Sep 2025*

- Built HR-management backend components and document-management infrastructure
- Implemented secure file-upload mechanisms with role-based access control (RBAC)
- Automated testing and deployment with GitHub Actions CI/CD pipelines
- Containerized backend services with Docker for consistent environments

R&D Engineering Intern *Telenity (R&D) | Jun 2024 – Aug 2024*

- Developed backend modules with Spring Boot
- Worked with Java, MongoDB, Kafka and SQL-based systems
- Focused on bug fixing, performance optimization and system stability

VOLUNTEERING EXPERIENCE

Educator & Co-leader *Anticverse Tech | 2022 – 2023*

- Helped manage the cybersecurity team and delivered training on data and web security

EDUCATION & CERTIFICATIONS

B.Sc. Computer Engineering Konya Food and Agriculture University | 2022 – 2026
Anthropic – Claude Academy Nov 2026: Model Context Protocol: Advanced Topics • Claude with Amazon Bedrock • Claude on Google Cloud • AI Fluency: Framework and Foundations
Mastering Model Mechanics Mercor | Nov 2026 • **Machine Learning Foundations** AWS Academy | Feb 2026
Social Engineering and Phishing BTK Akademi | Jan 2023

PROJECTS

Codebase RAG Q&A *Python • FastAPI • tree-sitter • pgvector*

- Repo Q&A with file:line citations via AST chunking, hybrid BM25 and vector retrieval, and an eval harness (94% hit@3)

Realtime Collab Editor *React • Yjs • WebSocket • Node.js*

- Google-Docs-style editor: Yjs CRDTs over a custom WebSocket server, live cursors, JWT sharing and version history

Distributed Rate Limiter *TypeScript • Redis • Lua • Prometheus*

- Token bucket, sliding and fixed window limiters on atomic Redis Lua scripts, so limits hold across replicas

Event-Driven Order System *TypeScript • RabbitMQ • OpenTelemetry • Docker*

- Five microservices running an order saga with compensating transactions, idempotent consumers and per-order tracing

Costsight — Cloud-Cost Anomaly Detection *Python • scikit-learn • Streamlit • AWS*

- Anomaly detection on AWS cost data with STL, Isolation Forest and Z-Score detectors, severity alerts and a dashboard

TECHNICAL SKILLS

Programming Languages: Java • Python • JavaScript • TypeScript • SQL • Verilog

Frontend: Next.js • React • Tailwind CSS

Backend: Node.js • Express.js • NestJS • Spring Boot • REST APIs

Security & Realtime: OWASP • Wireshark • Burp Suite • WebSocket • LiveKit

Databases: PostgreSQL • MongoDB • Redis • Prisma

DevOps & Tools: Docker • GitHub Actions • Kafka • RabbitMQ • Prometheus • OpenTelemetry • Linux • Postman

Cloud & AI: AWS • AWS ML Foundations • OpenAI API • FastAPI

Languages: English – C1 • Spanish – A1 • Turkish – Native